Section 12.6: Cylinders and Quadric Surfaces

Vectors and the Geometry of Space

MTH 310
Calculus III

Conic Sections Are Everywhere

Classifying Conics by Their Equation

$$ax^2 + by^2 + cx + dy + e = 0$$

Standard Forms of Conics

Standard Forms

- **Circle:** $(x-h)^2 + (y-k)^2 = r^2$
- **Ellipse:** $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
- **Hyperbola (horizontal):** $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$
- **Hyperbola (vertical):** $-\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

Example 1: Sketch $y^2 - 9x^2 = 1$

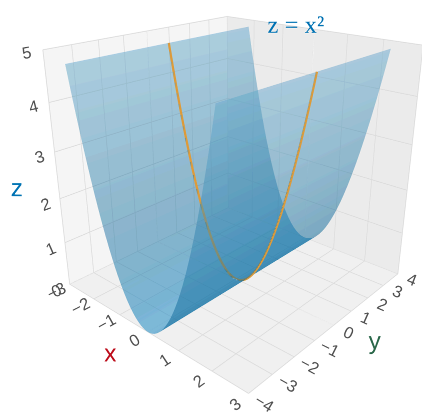
Example 2: Sketch $x^2 - 4y^2 = 1$

Example 3: Identify $x^2 - 4y^2 + 4x + 24y = 33$

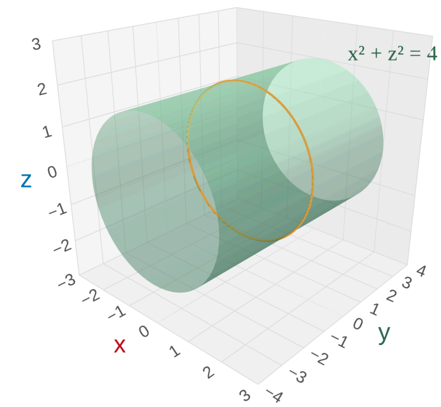
Cylinders in 3D

Cylinder

A **cylinder** is a surface formed by taking a 2D curve and extending it along the missing coordinate axis.
If one variable is absent from the equation, the graph is a cylinder that extends infinitely along that axis.



Interactive version



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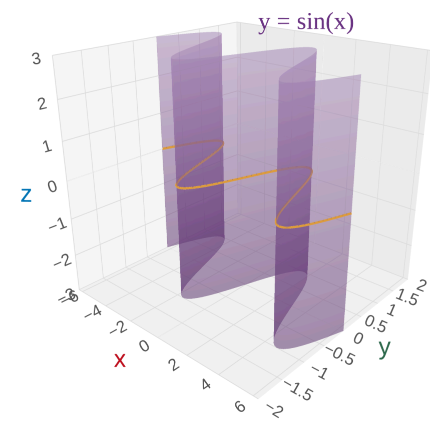

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Key idea: the missing variable is *free* — it can be anything.

Quadric Surfaces and Traces

Quadric Surface

A **quadric surface** is the 3D graph of a second-degree equation in x, y, and z.

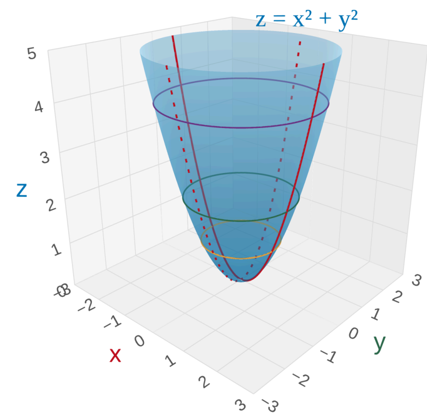
Trace

A **trace** is the cross-section you get by setting one variable equal to a constant.

- **Horizontal trace** ($z = k$): slice parallel to the xy-plane
- **Vertical trace** ($x = k$ or $y = k$): slice parallel to a coordinate wall

Strategy: Find traces in all three directions, then assemble the 3D shape.

Example 4: Identify $x^2 + y^2 - z = 0$



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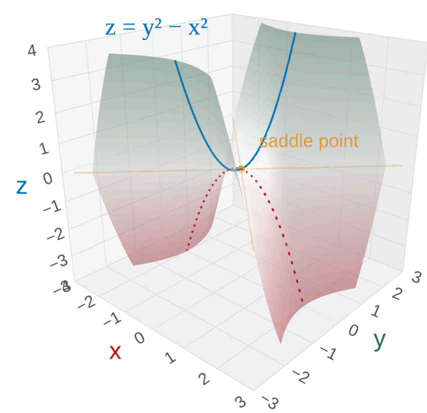
Elliptic paraboloid (bowl shape) — opens upward along the z-axis

Example 5: Identify $z = y^2 - x^2$



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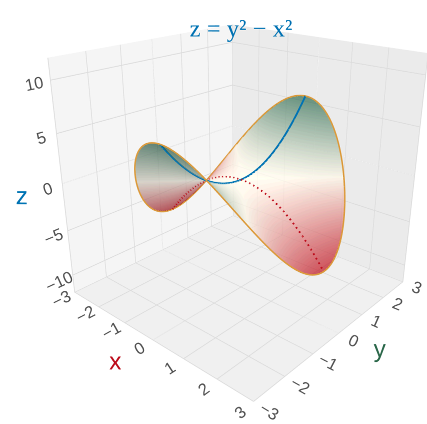
Hyperbolic paraboloid (saddle / "Pringle" shape)



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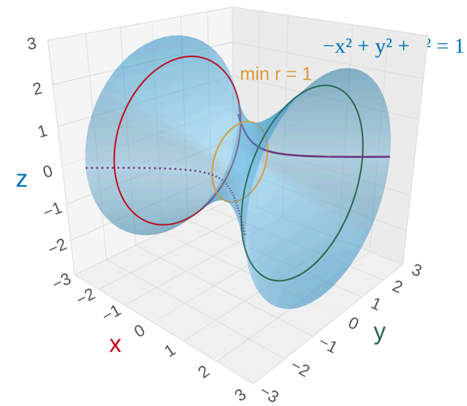
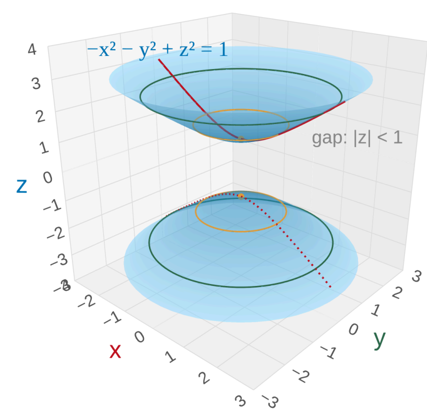
Example 6: Identify $-x^2 + y^2 + z^2 = 1$



Interactive version

Hyperboloid of one sheet — wraps continuously around the x-axis

Example 7: Identify $-x^2 - y^2 + z^2 = 1$



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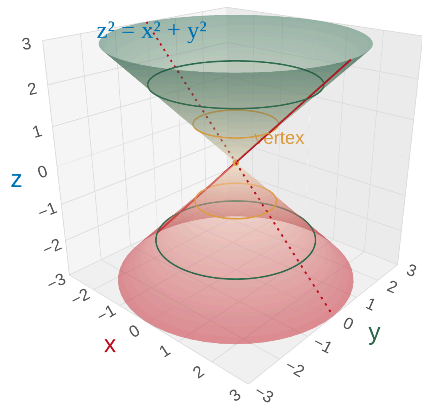
Hyperboloid of two sheets — two separate pieces with a gap between them

Example 8: Identify $z^2 = x^2 + y^2$



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Cone — the boundary case between hyperboloids of one and two sheets

Homework

Section 12.6: 1, 3, 6, 8, 13, 17, 21, 23–30, 47, 49